

4a	By Newton's Second Law, taking upwards as positive, $F_{\text{net}} = m a$ $F_E - m g = 0$ $q E = m g$ $q (\Delta V/d) = m g$ $\Delta V = m g d / q = (2.0 \times 10^{-14}) (9.81) (20 \times 10^{-3}) / (7.85 \times 10^{-18})$ $= 500 \text{ V}$ <i>Arriving at the correct number through incorrect physics does not yield any marks.</i>	M1 equation M1 numbers A1 answer
4bi	The potential difference is vertically applied. Hence, the electric field acts in the vertical direction. Only the vertical acceleration is affected. Since there is no change in the horizontal forces, horizontal motion is not affected. Hence, the time taken is still equal to 60 mm divided by 1.5 m s^{-1}. <i>Correct conclusions without a correct explanation are not awarded any marks.</i>	B1 B1
4bii	$2qE - mg = ma_y$ Since originally, the electric force qE was equal to the weight mg, $2mg - mg = ma_y$ $mg = ma_y$ $a_y = g$ <i>Students must explicitly and clearly show why $a = g = 9.81 \text{ m s}^{-2}$.</i> $v_x = 1.5 \text{ m s}^{-1}$ (unchanged) $t = s_x / v_x = (60 \times 10^{-3}) / (1.5) = 4.0 \times 10^{-2} \text{ s}$ $v_y = a_y t = (9.81) (4.0 \times 10^{-2}) = 0.3924 \text{ m s}^{-1}$ <i>Students must show, with appropriate formulae, how $v_y = 0.3924 \text{ m s}^{-1}$ is obtained.</i> $v = (v_x^2 + v_y^2)^{1/2} = 1.55 \text{ m s}^{-1}$ <i>Students must show, with an appropriate formula, how $v = 1.55 \text{ m s}^{-1}$ obtained.</i> <i>Arriving at the correct number by accident does not yield any marks.</i>	B1 B0 M1 M1

2021

HWA CHONG INSTITUTION (COLLEGE SECTION)

C2

5a

It is the energy converted per unit charge

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